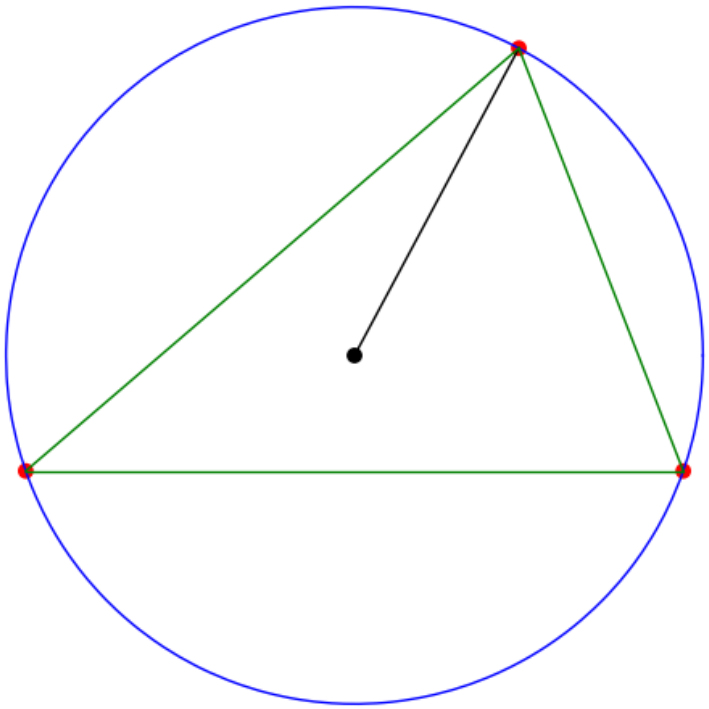
MCS 320 Project Three : Angles of Triangles in the Circumcircle
due Friday 20 November 2026 at 1pm

We apply SageMath to justify a geometric observation about the angles of triangles inside a circle.

1. The Circumcircle of a Triangle

Three points not on a line define a unique circle passing through those three points. We call this circle *the circumcircle of a triangle*. A general triangle is defined by the points $(0,0)$, $(x_1,0)$, and (x_2,y_2). An example for $x_1 = 0.56$, $x_2 = 0.42$ and $y_2 = 0.36$ is shown below.



The expressions of the coordinates for the center (c_x, c_y) and the radius r as function of (x_1, x_2, y_2) are

$$c_x = \frac{1}{2}x_1, \quad c_y = -\frac{x_1x_2 - x_2^2 - y_2^2}{2y_2}, \quad r = \frac{1}{2|y_2|}\sqrt{x_1^2x_2^2 - 2x_1x_2^3 + x_2^4 + x_1^2y_2^2 - 2x_1x_2y_2^2 + 2x_2^2y_2^2 + y_2^4}.$$

Observe the $|y_2|$ in the denominator for the radius: the closer the third point is to the base of the triangle, the larger the radius.

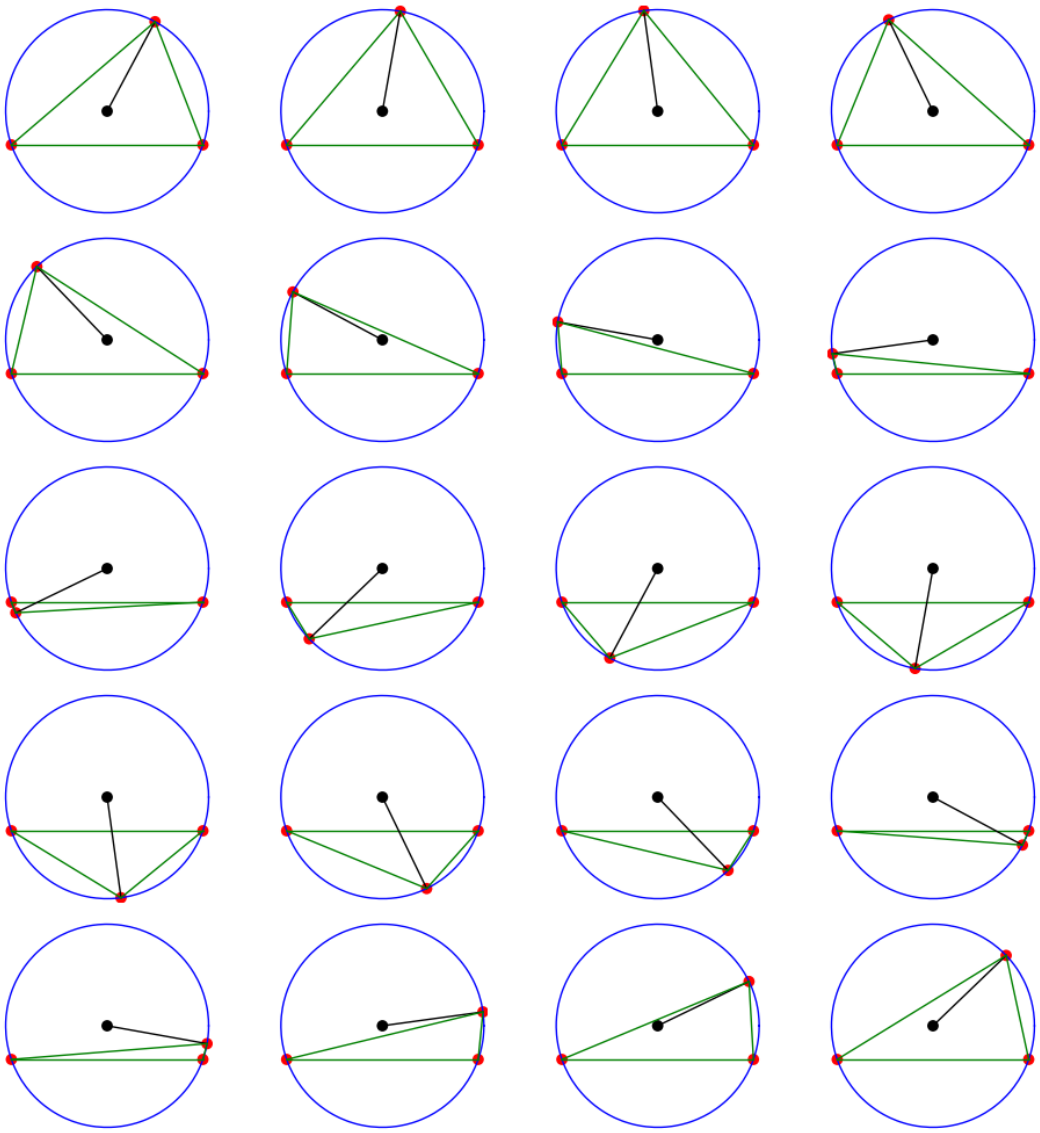
Assignment One. Derive the expressions for c_x, c_y, r, and make the above plot.

1. Set up the equations that express that $(0,0)$, $(x_1,0)$ and (x_2,y_2) are at the squared distance r from the center (c_x, c_y).
2. Compute a lexicographic Groebner basis for the variable order $c_x > c_y > r > x_1 > x_2 > y_2$.
3. From the Groebner basis, select those polynomials which can be solved for c_x, c_y, and r to give the above expressions. Then solve those polynomials to obtain the expressions.

Make the plot for $x_1 = 0.56$, $x_2 = 0.42$ and $y_2 = 0.36$.
In addition, make a plot for random positive values for x_1, x_2, and y_2.

2. An Animation

Keeping the base of the triangle fixed, we consider the triangles for the point (x_2, y_2) on the circumcircle. All frames of an animation starting at the configuration shown in the previous plot are displayed below.

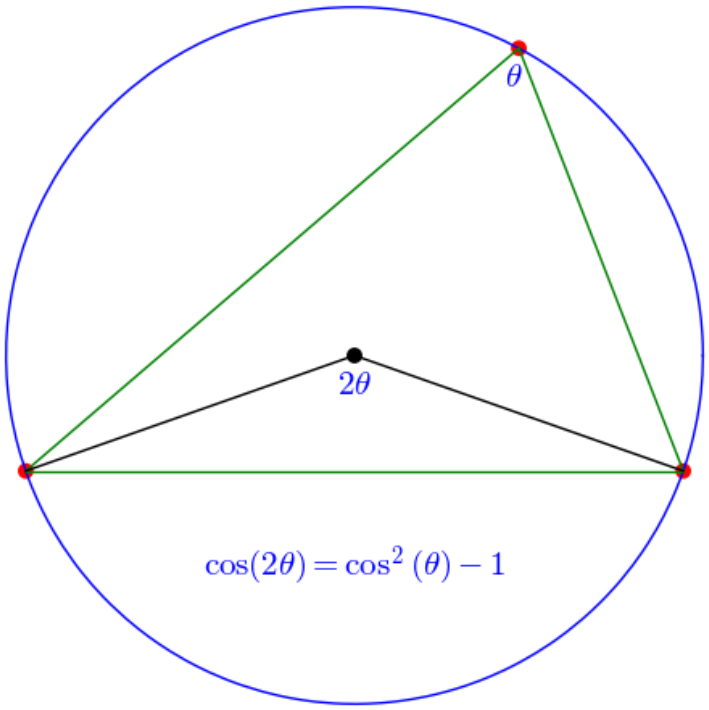


Assignment Two. Make an animation of 20 frames, using the values $x_1 = 0.56$, $x_2 = 0.42$, $y_2 = 0.36$, for the first frame, and then in polar coordinates for (x_2, y_2), use equidistant angles to define the frames.

3. Comparing Angles

Considering the plots in the animation, we could conjecture that if we look at the angle θ by the vectors originating at (x_2, y_2) and pointing to the two other points at the base of the triangle, that θ remains constant for all triangles defined by all points (x_2, y_2) on the same circle.

The idea to justify the conjecture is to relate the angle θ of the moving point to the fixed angle of the vectors originating at the center of the circle and pointing to the end points of the base of the triangle. This fixed angle at the center is 2θ.



Assignment Three.

1. Numerically, compute all 20 values for angle θ, as defined in the animation.
   Summarize your observations.
2. Symbolically, compute expressions for $\cos(2\theta)$ and $\cos^2(\theta)$,
   as functions of the three parameters x_1, x_2, and y_2.
3. After substitution of the expressions for $\cos(2\theta)$ and $\cos^2(\theta)$, show, numerically and symbolically that $\cos(2\theta) - \cos^2(\theta) + 1$ reduces to zero, therefore justifying that θ is fixed.

4. the Deadline is Friday 20 November 2026 at 1pm

Upload your solution to gradescope before 1pm on Friday 20 November 2026.

The solution consists of one single notebook, organized according to the assignments. Mark the start of each solution to an assignment using a heading in a cell. Your notebook should run from top to bottom as a program without errors. Apply proper formatting in your notebook so it reads like a technical report if you would print it. Document the execution cells with complete sentences, properly formatted in markdown cells.

You may (not must) work in pairs for this project. A pair consists of two, not three or more. If you decide to work in a pair, then you must send an email with the name of your partner and with the email address of your partner in the copy of the email to `janv@uic.edu`, before 5pm on Friday 13 November. If working in a pair, then only one Jupyter notebook should be submitted.

If you have questions or difficulties with the project, feel free to ask questions during office hour.